

Faster data

SDXC cards have a data transfer rate of 300 MBps and can go up to 2 TB

And some more

Transcend's JetFlash 600 pen drive has read speeds of 32 MBps and write speeds of 18 MBps



an external power connection is mandatory for the drive to work. In addition to this all the three drives were very impressive on the aesthetic aspect.

Silver bodied WD My Book Studio has an almost circular front panel with a pin hole which shows the power light. An impressive aspect about this product is the e-ink display on the front panel which gauges the capacity of the hard drive. It also displays other information like the name of the HDD, type of data and number of files. On the rear it has two firewire ports, a USB port and power button. This drive has vents on the top, rear and the bottom to allow heat dissipation. The metal body is quite sturdy.

WD's My Book Elite is very similar to My Book Studio in appearance and has vents on the top, rear and bottom side of the hard drive. This drive however does not have a Firewire port.

On connecting both the WD My Book Studio and WD My Book Elite, the WD SmartWare software starts automatically. This software allows the user to name the external drive and password protect it. It also has smart

HOW WE TESTED

We divided the external hard drives into two main parts based on the portability and size, namely Portable Hard Drives and Desktop Hard Drives. Portable Drives comprised compact drives most of which did not have an external power connector whereas Desktop drives comprised drives which were comparatively bigger in size and had an external power connector.

The hardware setup was made up of an ASUS P5Q3 motherboard with an Intel Core 2 QX9650 processor, 4 GB Corsair Dominator DDR3 RAM, 300GB WD Velociraptor hard disk drive and GeForce 7300LE graphics card. The power was supplied by Corsair HX620. This setup was maintained throughout the tests except for Buffalo 1 TB drive. While testing this drive only an interface card was added as this

Buffalo drive had a USB 3.0 interface unlike all the other drives which had USB 2.0 interfaces.

External drives are meant to give impressive read / write speeds when connected to the system, so the theoretical benchmarks used in the comparison tests were HD Tach and SiSoft Sandra 2010. HD Tach measures the performance of external drives by bypassing as many software as possible. This helps in giving read/write speeds, CPU utilization and random access times as well as interface burst speed. Both the HD Tach and SiSoft Sandra tests require an unformatted drive to perform the Write testing. After finishing with these two benchmarks we formatted and divided the external hard drives into two partitions of equal size. All the benchmark processes were

run on the minimum Windows Vista settings. - All benchmarks were run on Windows Vista with system restore and paging file fixed to 4 GB.

In data transfer tests we followed three test processes. File Writing for a 4 GB sequential and assorted file; File reading for a 4 GB sequential and assorted file and a Photoshop image loading. To simulate a write test we copied files from the HDD to the external drive and vice versa. To prevent giving drives an undue advantage while reading due to the caching process, we rebooted the system after completing the Write and then the Read operations. Adobe Photoshop CS2 was used to test the amount of time taken to load a 1GB image. The scratch disk was placed on the external drive.

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Backup and Retrieve options. A very interesting feature is the backup option. On clicking on the Backup tab, we get a graphical representation of the available data on our system drives (default C:) divided into 6 options like system, pictures, documents, music, movies and others. For quick backups, the user just has to select the desired option and click Run Backup. The transfer starts which can be seen visually. However, there is no option to select individual folders for this purpose.

FreeAgent Desk, the 1 TB external hard drive from Sea-

gate comes in a silver brush metal finish with a stand. It has a power input and a mini USB connection on the rear end.

The Buffalo DriveStation 1 TB hard drive, was the only one with a USB 3.0 interface (with almost triple data transfer speeds as compared to USB 2.0). It comes in a black glossy body. A silver line runs through the front panel with a blue light on top indicating power in the USB 3.0 mode and a green glow in USB 2.0 mode.

We used the Buffalo USB 3.0 interface card for PCI Express to conduct the test. It has a vent on the rear side to allow heat dis-

sipation. The glossy finish looks smashing, though it tends to collect a lot of dust and definitely not suited to Indian conditions.

Performance

The external hard drives have a buffer size that is much greater than the portable hard drives (32 MB as opposed to 8 MB of portable hard drives), as a result of which one gets a much better performance. But this performance comes at the cost of portability.

WD's My Book Elite gave a marginally better performance with average read and write speeds. But when the WD My



Buffalo DriveStation